Total GI

COMPREHENSIVE STOOL PROFILE

PATIENT INFORMATION		PROVIDER INFORMATION		SAMPLE INFORMATION	
Name	: John Doe	Name	: Jane Doe	Order #	: 202610307851
DOB	: 01/01/2000	NPI Number	: 1234567890	Specimen	: Stool
Gender	: Male	Credentials	: MD	Collected	: 08/16/2026
				Sample Received	: 08/18/2026
				Result Reported	: 09/04/2026

SUMMARY

PATHOGENS DETECTED BY MULTIPLEX PCR

PATHOGENIC BACTERIA
Yersinia enterocolitica
FUNGUS
Candida albicans
PARASITES
Cryptosporidium

PARASITES DETECTED BY MICROSCOPY

MALABSORPTIVE / OSMOTIC DIARRHEA-CAUSING
Cryptosporidium spp.
CYST
Cryptosporidium (Oocyst)

ABNORMALITIES IN STOOL EXAMINATION

MICROSCOPIC APPEARANCE	RESULT	
KOH for Yeast	Few	◉

◉ Represents abnormal microscopic and macroscopic appearance

ABNORMALITIES INTESTINAL HEALTH MARKERS (EIA)

TEST NAME	RESULT	
Fecal Secretory IgA*	2235	↑

↑ Represents elevated level from reference range ↓ Represents lessened level from reference range

PATHOGENS DETECTED BY CULTURE

PATHOGENIC BACTERIA	RESULT	
Yersinia enterocolitica	+4	
FUNGUS	**RESULT**	
Candida albicans	+4	

Shows microbial growth by quadrant streak method

RSM Diagnostics Lab (CLIA ID #: 08D2253750)
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STOOL EXAMINATION

Order : 202610307851
Patient : John Doe
DOB : 01/01/2000

MACROSCOPIC

MACROSCOPIC APPEARANCE	RESULT	FLAG	REFERENCE INTERVAL
Color	Brown	-	Brown
Consistency	Soft	-	Soft
pH	6	-	5.80-7.00
Occult Blood	Negative	-	Negative
Mucus	Negative	-	Negative

MICROSCOPIC

MICROSCOPIC APPEARANCE	RESULT	FLAG	REFERENCE INTERVAL
WBC	Negative	-	Negative
RBC	Negative	-	Negative
Charcot-Leyden Crystals	Negative	-	Negative
Fat Stain	Negative	-	Negative
KOH for Yeast	Few	◉	RARE: 1-2 Per slide FEW: 2-5 Per slide MODERATE: 5-10 Per HPF MANY: >10 Per HPF High Power Field (HPF)
Carbohydrates	Negative	-	Negative

COLOR : Brown is the typical color of regular stool, while deviations in color may suggest abnormal gastrointestinal conditions.

CONSISTENCY : A formed stool is considered normal. Variations to this may indicate abnormal gastrointestinal tract conditions.

MUCOUS : Mucous production may indicate the presence of an infection, inflammation, or malignancy.

BLOOD : The presence of blood in the stool may indicate a possible gastrointestinal tract ulcer and must always be investigated immediately.

FECAL FAT : Fecal fat or steatorrhea is indicative of malabsorption in the digestive system. Conditions such as pancreatic insufficiency, celiac disease, Crohn's disease, and other disorders.

STOOL CARBOHYDRATE : Measures the presence of unabsorbed carbohydrates in stool. Elevated levels may be associated with impaired carbohydrate digestion or absorption and can occur in conditions affecting intestinal enzyme activity, mucosal integrity, or gastrointestinal transit time. Results should be interpreted in conjunction with clinical findings and other laboratory data.

INTESTINAL MARKERS

Order : 202610307851
Patient : John Doe
DOB : 01/01/2000

TEST NAME	RESULT	FLAG	REFERENCE INTERVAL
◆ GUT INFLAMMATION MARKERS			
Fecal Calprotectin*	170	-	<173 ug/mL
◆ GUT INTESTINAL INSUFFICIENCY			
Fecal Pancreatic Elastase*	643	-	>200 ng/ml
◆ GUT IMMUNE RESPONSE			
Fecal Secretory IgA*	2235	↑	510-2040 ug/ml
◆ INTESTINAL PERMEABILITY			
Fecal Zonulin *	53	-	0.25-64.00 ng/ml
◆ OTHER GUT HEALTH MARKER			
Fecal Anti-Gliadin IgA*	74	-	<120 U/L
◆ STOOL ANTIGEN			
H. pylori Fecal Ag, EIA*	Negative		Negative

↑ Represents elevated level from the reference range

↓ Represents lessened level from the reference range

* The analytical performance characteristics of this test have been established and validated by RSM Diagnostics Lab in accordance with applicable CLIA regulations for clinical laboratory testing. This test has not been cleared or approved by the U.S. Food and Drug Administration (FDA).

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GI MULTIPLEX PCR

Order : 202610307851
Patient : John Doe
DOB : 01/01/2000

◆ COMMENSAL BACTERIA

BACTERIA	RESULT		RANGE
Akkermansia muciniphila	1E6		1E3-1E12
Bacteroides fragilis	2E5		1E3-1E12
Enterococcus faecalis	2E5		1E3-1E12
Enterococcus faecium	<DL		1E3-1E12
Fusobacterium	1E5	L	1E3-1E12
Faecalibacterium prausnitzii	<DL		1E3-1E12
Roseburia spp.	1E6		1E3-1E12

◆ COMMENSAL INFLAMMATORY & AUTOIMMUNE BACTERIA

BACTERIA	RESULT		RANGE
Fusobacterium spp.	1E5	L	1E3-1E12
Prevotella spp.	1E6		1E3-1E12
Clostridium perfringens	Not Detected		1E3-1E12
Sutterella spp.	Not Detected		1E3-1E12

◆ COMMENSAL OVERGROWTH BACTERIA

BACTERIA	RESULT		RANGE
Desulfovibrio spp.	2E5		1E3-1E12
Methanobacteriaceae (Family)	<DL		1E3-1E12

• The results are shown using scientific notation, where the letter “E” represents the exponent (e.g., 1.2E6 means 1.2 × 106 or 1,200,000). The values from 1E3 to 1E12 are the test's detection range.

• Not Detected/ND means the pathogen is not detected and is shown as a gray-shaded area with no indicator. “< DL” means the pathogen is present but below the detection limit. In this case, the indicator points to the start of the graph (1E3) without any color change.

• The bar graph also uses colors to show levels of commensal bacteria: yellow for low and green for high, where a result value >=1E3 to <=1E5 represents low and a value>1E5 to <=1E12 represents high.

◆ VIRUSES

VIRUSES	RESULT
Enterovirus	Not Detected
Human adenovirus (HAdV)	Not Detected
Human astrovirus (HAtV)	Not Detected
Human norovirus (HNoV) G1	Not Detected
Human norovirus (HNoV) G2	Not Detected
Human rotavirus (HRTV)	Not Detected
Human sapovirus (HSaV)	Not Detected

◆ PATHOGENIC BACTERIA

BACTERIA	RESULT	RANGE
Campylobacter	Not Detected	<1E3
Clostridium difficile	Not Detected	<1E3
Enterohemorrhagic/verotoxin producing Escherichia coli (EHEC)	Not Detected	<1E3
Enteroinvasive Escherichia coli (EIEC)	Not Detected	<1E3
Salmonella enterica	Not Detected	<1E3
Shiga-like Toxin E. coli stx1	Not Detected	<1E3
Shiga-like Toxin E. coli stx2	Not Detected	<1E3
Yersinia enterocolitica	Detected	<1E3
Helicobacter pylori	Not Detected	<1E3

◆ FUNGUS

FUNGUS	RESULT	RANGE
Candida (pan-candida)	Not Detected	<1E3
Candida albicans	Detected	<1E3
Candida glabrata	Not Detected	<1E3

◆ **PARASITES**

PARASITES	RESULT	RANGE
Cryptosporidium	Detected	<1E3
Entamoeba histolytica	Not Detected	<1E3
Giardia lamblia	Not Detected	<1E3

◆ **DYSBIOTIC & OVERGROWTH**

BACTERIA	RESULT	RANGE
Enterococcus faecalis	Detected	<1E3
Enterococcus faecium	Not Detected	<1E3
Morganella morganii	Not Detected	<1E3
Pseudomonas aeruginosa	Not Detected	<1E3
Staphylococcus aureus	Not Detected	<1E3
Streptococcus agalactiae (Group B)	Not Detected	<1E3
Streptococcus pneumoniae	Not Detected	<1E3
Streptococcus pyogenes	Not Detected	<1E3

◆ **INFLAMMATORY & AUTOIMMUNE BACTERIA**

BACTERIA	RESULT	RANGE
Acinetobacter baumannii	Not Detected	<1E3
Citrobacter koseri	Not Detected	<1E3
Citrobacter freundii	Not Detected	<1E3
Klebsiella pneumoniae,oxytoca	Detected	<1E3
Proteus mirabilis	Not Detected	<1E3
Mycobacterium avium	Not Detected	<1E3
Mycoplasma pneumonia	Not Detected	<1E3
Chlamydia trachomatis	Not Detected	<1E3

ANTIBIOTIC RESISTANCE GENES

◆ B-LACTAMS

RESISTANCE GENES	RESULT	RESISTANCE GENES	RESULT
ACT-1/3/2/34	Not Detected	AMPC	Detected
CTX-M 1	Not Detected	CTX-M 2	Not Detected
CTX-M 8	Not Detected	CTX-M 9	Not Detected
OXA-1	Not Detected	FOX-17	Not Detected
IMP-1	Not Detected	IMP-2	Not Detected
VIM-1	Not Detected	VIM-2	Not Detected
NDM-1	Not Detected		

◆ MACROLIDES

RESISTANCE GENES	RESULT
ErmA	Detected
ErmB	Not Detected
ErmC	Not Detected

◆ FLUOROQUINOLONES

RESISTANCE GENES	RESULT
Quinolone A	Not Detected
Quinolone B	Not Detected

◆ SULFONAMIDES

RESISTANCE GENES	RESULT
Sulfamethoxazole 1 (Sul1)	Not Detected
Sulfamethoxazole 1 (Sul2)	Not Detected

◆ TETRACYCLINE

RESISTANCE GENES	RESULT
Tetracycline B	Not Detected
Tetracycline M	Detected

◆ METHICILLIN

RESISTANCE GENES	RESULT
MECA	Not Detected

◆ TRIMETHOPRIM

RESISTANCE GENES	RESULT
DFRA1	Not Detected
DFRA5	Not Detected

◆ VANCOMYCIN

RESISTANCE GENES	RESULT
Vancomycin AB	Not Detected

GI PARASITOLOGY MICROSCOPIC

Order : 202610307851
Patient : John Doe
DOB : 01/01/2000

GI PARASITOLOGY

◆ NORMAL FLORA / COMMENSAL

PARASITES	RESULT
Blastocystis hominis	—
Endolimax nana	—
Chilomastix mesnili	—
Entamoeba coli cyst	—
Entamoeba coli	—

◆ DIRECT MUCOSAL INVADERS & INFLAMMATORY

PARASITES	RESULT
Entamoeba histolytica	—
Entamoeba histolytica cyst	—
Balantidium coli	—
Strongyloides stercoralis	—

◆ MAST CELL ACTIVATION & IMMUNE MEDIATED

PARASITES	RESULT
Dientamoeba fragilis troph	—
Dientamoeba fragilis Cyst	—
Giardia lamblia	—
Blastocystis	—

◆ LARGE-BURDEN & LUMINAL OBSTRUCTION

PARASITES	RESULT
Ascaris lumbricoides ova	—
Cystoisopora spp.	—
Cyclospora cayetanensis	—
Hymenolepis nana	—

◆ MALABSORPTIVE / OSMOTIC DIARRHEA-CAUSING

PARASITES	RESULT
Giardia lamblia cyst	—
Cryptosporidium spp.	⚠

◆ MUCOSAL NIPPERS & BLOOD FEEDERS

PARASITES	RESULT
Ancylostoma duodenale	—
Necator americanus	—
Hookwarm ova	—
Trichuris trichiura ova	—

◆ AUTOINFECTION & HYPERINFECTION RISK

PARASITES	RESULT
Strongyloides Stercoralis	—

◆ CYST

CYST	RESULT	CYST	RESULT
Acanthamoeba polyphaga cyst	—	Cryptosporidium (Oocyst)	⚠
Dientamoeba fragilis cyst	—	Toxoplasma gondii oocyst	—
Entamoeba nana cyst	—	Gongylonemia pulchrur	—
Isospora belli sporocyst	—		

⚠ Detected — Not Detected

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GI PARASITOLOGY MICROSCOPIC

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◆ OVA

OVA	RESULT	OVA	RESULT
Capillaria hepatica ova	—	Diphyllobothrium latum ova	—
Entamoeba hortense ova	—	Entamoeba pancreaticum ova	—
Enterobius vermicularis ova	—	Fasciola hepatica ova	—
Fasciolopsis buski ova	—	Hymenolepis diminuta ova	—
Theileria orientalis ova	—	Hymenolepis nana ova	—
Iodamoeba buetschlii	—	Metagonimus yokogawai ova	—
Opisthorchis viverrini ova	—	Schistosoma japonicum ova	—
Schistosoma mansoni ova	—	Taenia ova	—

◆ PARASITES

PARASITES	RESULT	PARASITES	RESULT
Acanthamoeba trophozoite	—	Capillaria philippinensis	—
Chilomastix mesnili	—	Chylomastix hominis	—
Chylomastix mednili	—	Clonorchis sinensis	—
Cryptosporidium spp.	—	Diphyllobothrium latum	—
Dipylidium caninum	—	Entamoeba cell	—
Entamoeba cell trophozoite	—	Enterobius vermicularis	—
Iodamoeba buetschlii	—	Isospora	—
Metagonimus yokogawai	—	Paragonimus spp.	—
Sarcocystis	—	Theileria orientalis	—
Toxoplasma gondii	—	Trichinella spirali	—
Trichomonads	—	Trichomonas vaginalis	—

⚠ Detected — Not Detected

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BACTERIOLOGY CULTURE & SENSITIVITY

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◆ BENEFICIAL FLORA

MICROBES	RESULT	GROWTH
Bacteroides Family	+4	
Bifidobacterium Family	+4	
Clostridial Family	+4	
Lactobacillus Family	+2	

◆ OPPORTUNISTIC/OVERGROWTH BACTERIA*

MICROBES	RESULT	GROWTH
Klebsiella pneumoniae	+3	

◆ PATHOGENIC BACTERIA

MICROBES	RESULT	GROWTH
Campylobacter spp.	No Growth	
Enterohemorrhagic E. coli (EHEC) - O157:H7	No Growth	
Salmonella spp.	No Growth	
Shigella spp.	No Growth	
Yersinia enterocolitica	+4	

No growth detected

Shows the growth of beneficial flora by quadrant streak method

Shows bacterial pathogens growth by the quadrant streak method

* **Opportunistic / Overgrowth Bacteria: Not reporting antimicrobial sensitivity**

BACTERIOLOGY CULTURE & SENSITIVITY

Order : 202610307851
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◆ ANTIMICROBIAL SENSITIVITY

ANTIBIOTICS	RESULT	
Ampicillin	Resistant	
Amoxicillin	Resistant	
Doxycycline	Sensitive	
Erythromycin	Resistant	
Penicillin	Resistant	
Trimethoprim/Sulfamethoxazole	Sensitive	

◆ NATURAL AGENT SENSITIVITY

ANTIBIOTICS	RESULT	
Alchornea	No	
Argentyn silver	No	
Clove	Low	
Cryptolepis	No	
Caprylic Acid	No	
Garlic	Low	
Juniper	No	
Oil of Oregano	High	
Oregon Grape	No	
Red Root	No	

Antibiotic - Sensitive against the microbes **Antibiotic - Resistant against the microbes**

In this assay, inhibition is defined as the reduction in organism growth resulting from exposure to a test substance. The level of inhibition indicates how effectively the substance limits the growth of the organism under in vitro conditions. High inhibition indicates a greater ability of the substance to limit organism growth, whereas low inhibition indicates a lesser ability. The evaluated natural products should be considered investigational and should not be regarded as standard clinical treatments or approved therapeutic agents.

MYCOLOGY CULTURE & SENSITIVITY

Order : 202610307851
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DOB : 01/01/2000

◆ FUNGAL CULTURE

MICROBES	RESULT	GROWTH
Candida albicans	+4	

◆ ANTIFUNGAL SENSITIVITY

ANTIFUNGAL	RESULT	
Amphotericin-B	Resistant	
Clotrimazole	Sensitive	
Fluconazole	Sensitive	
Itraconazole	Resistant	
Ketoconazole	Resistant	
Nystatin	Sensitive	
Voriconazole	Resistant	

◆ NATURAL AGENT SENSITIVITY

ANTIFUNGAL	RESULT	
Alchornea	Low	
Argentyn silver	No	
Clove	Low	
Cryptolepis	No	
Caprylic Acid	No	
Garlic	High	
Juniper	No	
Oil of Oregano	High	
Oregon Grape	Low	
Red Root	Low	

Represents the antifungal is sensitive against the microbes

Represents the antifungal is resistant against the microbes

Shows microbial growth by quadrant streak method

In this assay, inhibition is defined as the reduction in organism growth resulting from exposure to a test substance. The level of inhibition indicates how effectively the substance limits the growth of the organism under in vitro conditions. High inhibition indicates a greater ability of the substance to limit organism growth, whereas low inhibition indicates a lesser ability. The evaluated natural products should be considered investigational and should not be regarded as standard clinical treatments or approved therapeutic agents.

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INTESTINAL MARKERS

CALPROTECTIN

- Calprotectin is a protein found in white blood cells, particularly in neutrophils. Calprotectin becomes available in the intestinal lumen via leukocyte shedding, active secretion, cell disturbance, and cell death. Elevated levels of calprotectin in stool samples can indicate inflammation in the intestines. Fecal calprotectin is associated with conditions such as inflammatory bowel disease (IBD).

PANCREATIC ELASTASE

- Elastase is an enzyme produced by the pancreas that helps break down proteins, fats and carbohydrates. It plays a crucial role in the digestion process by breaking down dietary proteins into smaller peptides and amino acids, which can be absorbed by the body.
- Low levels of elastase in the stool may suggest pancreatic insufficiency. Testing for elastase levels in stool samples is a noninvasive method to assess pancreatic function.

ANTI GLIADIN ANTIBODIES

- Anti Gliadin antibodies directed against gliadin, a component of gluten found in wheat and other related grains such as barley and rye. These antibodies are relevant in the context of gluten related disorders, particularly celiac disease. May indicate an immune response to gluten, but not specific to celiac disease. If positive further testing is advised.

SECRETARY IGA

- Secretory IgA (sIgA) is a type of immunoglobulin A that plays a critical role in the immune function of mucosal membranes. sIgA is predominantly found in mucosal secretions such as saliva, tears, colostrum, and gastrointestinal, respiratory, and urogenital tracts. It is the first line of defense against pathogens.
- IgA protects against infections in the mucosal surface, Low levels of sIgA are associated with increased susceptibility to infections in the GI tract.
- Measurement of sIgA levels can be useful in diagnosing certain immunodeficiencies and monitoring mucosal immunity.

ZONULIN

- Zonulin is a protein that plays a critical role in regulating intestinal permeability. It is involved in the modulation of tight junctions between the cells lining the intestinal wall.
- Elevated zonulin levels can indicate increased gut permeability, which may lead further investigation and management of underlying conditions. Elevated zonulin may be associated with autoimmune disease, inflammatory bowel disease, celiac disease, and metabolic disease.
- Zonulin release may be triggered by bacteria, gluten and certain dietary compounds.

FECAL EOSINOPHIL ACTIVATION PROTEIN (EDN)

- Fecal Eosinophil-Derived Neurotoxin (EDN) is a marker associated with eosinophil activation in the gastrointestinal tract and may reflect eosinophilic or allergic-type intestinal inflammation. Elevated levels can be seen in certain inflammatory, allergic, or immune-mediated conditions affecting the bowel. **Note:** Results should be interpreted in conjunction with clinical findings and other laboratory data.

STOOL HELICOBACTER PYLORI

- Stool Helicobacter pylori Antigen is a test used to detect antigens associated with Helicobacter pylori infection in stool.
- A positive result may indicate the presence of H. pylori, which has been associated with certain gastrointestinal conditions.
- Results should be interpreted in conjunction with clinical findings and other laboratory data.

NORMAL FLORA / COMMENSAL

- Certain parasites identified may be classified as commensal or non-pathogenic parasites, commonly present in the gastrointestinal tract..

MAST CELL ACTIVATION & IMMUNE-MEDIATED

- Certain intestinal parasites can stimulate the immune system, leading to activation of mast cells as part of the body's mechanism. (e.g., increased intestinal permeability, histamine release).
- Parasite antigens may provoke mast cells to release mediators such as histamine, cytokines, and proteases, contributing to localized or systemic inflammation.

DIRECT MUCOSAL INVADERS & INFLAMMATORY

- Certain parasites possess the ability to invade or attach to intestinal mucosa, resulting in direct epithelial disruption.
- Mucosal invasion may lead to localized tissue injury, ulceration, or compromised barrier integrity.
- This interaction can trigger a host inflammatory response, including recruitment of immune cells and release of pro-inflammatory mediators.

MALABSORPTIVE / OSMOTIC DIARRHEA-CAUSING

- Certain intestinal parasites may interfere with normal nutrient absorption, leading to malabsorption syndromes and altered gastrointestinal function.
- These organisms can disrupt the mucosal surface, impairing enzymatic activity and reducing nutrient uptake.
- Infection may result in osmotic or malabsorptive diarrhea, depending on the mechanism of intestinal disruption.

LARGE-BURDEN & LUMINAL OBSTRUCTION

- Certain parasitic infections may present with high organism burden within the intestinal lumen, potentially contributing to partial or functional obstruction.
- Luminal crowding and aggregation of organisms may contribute to mechanical blockage or impaired passage of intestinal contents.

MUCOSAL "NIPPERS" & BLOOD FEEDERS

- Certain organisms may exhibit mucosal adherence with tissue or blood-feeding behavior.
- Potential for localized epithelial injury and microhemorrhage.
- May be associated with chronic inflammatory response and iron depletion in prolonged cases.
- Clinical correlation recommended with symptoms such as fatigue, anemia, or gastrointestinal discomfort.
- Further evaluation may be warranted based on symptom severity and laboratory correlation (e.g., hemoglobin, ferritin, occult blood).

AUTOINFECTION & HYPERINFECTION RISK

- Certain parasites possess the ability to complete their life cycle within the host, leading to autoinfection and potential persistence without external re-exposure.
- Hyperinfection may lead to dissemination beyond the gastrointestinal tract, with potential involvement of pulmonary or systemic sites.

Total GI

COMPREHENSIVE STOOL PROFILE

Disclaimer

The assays are laboratory developed tests and/or the performance characteristics determined by RSM Diagnostics Lab. RSM Diagnostics Lab is a CLIA approved laboratory. Test results should be interpreted by a healthcare professional who can integrate these findings with clinical symptoms, medical history, and other diagnostic tests. Self-diagnosis and treatment based solely on laboratory test results can be harmful.

RSM DIAGNOSTICS LAB

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